第一章 热力学的基本概念和规律

1.0 $f(x,y,z)=0$

(1) $\left(\frac{\partial z}{\partial y}\right)_x = \frac{1}{\left(\frac{\partial y}{\partial z}\right)_x}$

(2) $\left(\frac{\partial z}{\partial y}\right)_x \left(\frac{\partial y}{\partial x}\right)_z \left(\frac{\partial x}{\partial z}\right)_y = -1$

证：引：$\left(\frac{\partial x}{\partial y}\right)_z = -\frac{\left(\frac{\partial f}{\partial y}\right)_{x,z}}{\left(\frac{\partial f}{\partial x}\right)_{y,z}}$

证：$df = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy + \frac{\partial f}{\partial z}dz = 0$

$dz=0$ 时：$\left(\frac{\partial x}{\partial y}\right)_z = \frac{dx}{dy} = -\frac{\frac{\partial f}{\partial y}}{\frac{\partial f}{\partial x}}$ D.

$\left(\frac{\partial z}{\partial y}\right)_x \cdot \left(\frac{\partial y}{\partial x}\right)_z \cdot \left(\frac{\partial x}{\partial z}\right)_y = (-1)^3 \cdot \frac{\frac{\partial f}{\partial y}}{\frac{\partial f}{\partial z}} \cdot \frac{\frac{\partial f}{\partial x}}{\frac{\partial f}{\partial y}} \cdot \frac{\frac{\partial f}{\partial z}}{\frac{\partial f}{\partial x}} = -1$ D.

(3) $\left(\frac{\partial y}{\partial x}\right)_z = -\frac{\left(\frac{\partial z}{\partial x}\right)_y}{\left(\frac{\partial z}{\partial y}\right)_x}$

证：$LHS = -\frac{\left(\frac{\partial f}{\partial x}\right)_{y,z}}{\left(\frac{\partial f}{\partial y}\right)_{x,z}}$，$RHS = -\frac{-\frac{\frac{\partial f}{\partial x}}{\frac{\partial f}{\partial z}}}{-\frac{\frac{\partial f}{\partial y}}{\frac{\partial f}{\partial z}}} = -\frac{\frac{\partial f}{\partial x}}{\frac{\partial f}{\partial y}}$

(4) $\left(\frac{\partial w}{\partial y}\right)_x = \left(\frac{\partial w}{\partial z}\right)_x \left(\frac{\partial z}{\partial y}\right)_x$，其中 $w=w(x,z)$

对于 $y=y(x,w(x,z))$，

$dy|_x = \frac{\partial y}{\partial x}dx + \frac{\partial y}{\partial w}dw = \frac{\partial y}{\partial w}dw$

$dw|_x = \frac{\partial w}{\partial x}dx + \frac{\partial w}{\partial z}dz = \frac{\partial w}{\partial z}dz$

$\to \left(\frac{\partial y}{\partial z}\right)_x = \frac{dy|_x}{dz|_x} = \left(\frac{\partial y}{\partial w}\right)_x \left(\frac{\partial w}{\partial z}\right)_x = \frac{\left(\frac{\partial w}{\partial z}\right)_x}{\left(\frac{\partial w}{\partial y}\right)_x}$ D.

(5) $\left(\frac{\partial w}{\partial x}\right)_z = \left(\frac{\partial w}{\partial x}\right)_y + \left(\frac{\partial w}{\partial y}\right)_x \left(\frac{\partial y}{\partial x}\right)_z$，其中 $w=w(x,y)$

证：$dz=0$ 时：

$dw = \left(\frac{\partial w}{\partial x}\right)_y dx + \left(\frac{\partial w}{\partial y}\right)_x dy$

$dy = d[y(x,z)] = \left(\frac{\partial y}{\partial x}\right)_z dx + \left(\frac{\partial y}{\partial z}\right)_x dz = \left(\frac{\partial y}{\partial x}\right)_z dx$

$\left(\frac{\partial w}{\partial x}\right)_z = \left(\frac{\partial w}{\partial x}\right)_y + \left(\frac{\partial w}{\partial y}\right)_x \left(\frac{\partial y}{\partial x}\right)_x$

1.1

热力学系统：微观粒子 -> 宏观体系

↘ 孤立系统：无物质、能量交换

闭系　　无物质，有能量

开系　　有～，有～

1.2

平衡态：宏观性质不随时间变化 ≠ 定态 e.g. DC Transport

自由度：状态参量的数目，几何 / 力学 / 化学 / 电磁；广延 / 强度

$w = f(x_1, x_2 \cdots x_n)$　　$\propto$ mol

1.3

热力学第零定律：$A \overset{\text{平}}{\underset{\text{衡}}{=}} B$，$B = C$ => $A = C$

温度：物体处于热平衡时具有相同的冷热程度。

1.4.

物态方程：温度 ~ 状态参量 函数关系

$T = \varphi(x_1, x_2 \sim x_n)$

$g(x_1, x_2 \cdots x_n, T) = 0$

(1) 体胀系数：$\alpha = \frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_P$

(2) 压强系数：$\beta = \frac{1}{P}\left(\frac{\partial P}{\partial T}\right)_V$

(3) 压缩系数：$\kappa = -\frac{1}{V}\left(\frac{\partial V}{\partial P}\right)_T$

$\left(\frac{\partial V}{\partial P}\right)_T \cdot \left(\frac{\partial P}{\partial T}\right)_V \cdot \left(\frac{\partial T}{\partial V}\right)_P = -1$

=> $\alpha = P\beta\kappa$

典型物态方程：

(1) 理想气体：$PV = NRT$

(2) Van der Waals 气体：$\left(P + \frac{aN^2}{V^2}\right)(V - Nb) = NRT$

(3) Onnes 方程：$P = \frac{NRT}{V}\left[1 + \frac{N}{V}B(T) + \left(\frac{N}{V}\right)^2 C(T) + \cdots\right]$

$B(T)$，$C(T)$ 为第二、第三位力系数

(4) 顺磁性固体：$\vec{m} = \frac{C\vec{H}}{T}$

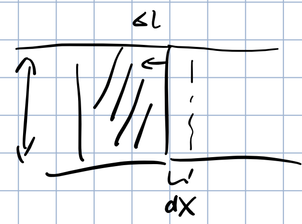
1.5.

准静态过程：缓慢、过程中时刻处于平衡态。

外界对系统作功：$đW = -P\,dV$

（静流体） $W = -\int_{V_A}^{V_B} P\,dV$

液体表面薄膜：$đW = 2\sigma l\,dx = 2\sigma\,dA$



电介质极化：$đW = V\vec{E}\cdot d\vec{D}$ $\quad \vec{D} = \varepsilon_0\vec{E} + \vec{P}$

$= V\vec{E}\cdot(\varepsilon_0\,d\vec{E} + d\vec{P})$

$= \underbrace{V\cdot d(\frac{1}{2}\varepsilon_0\vec{E}^2)}_{\text{激发电场的功}} + \underbrace{V\vec{E}\cdot d\vec{P}}_{\text{介质极化的功（外界电场与电偶极作用）}}$

磁介质极化：$đW = V\vec{H}\cdot d\vec{B}$ $\quad \vec{B} = \mu_0\vec{H} + \vec{m}$

$= V\vec{H}\cdot(\mu_0\,d\vec{H} + d\vec{m})$

$= V\cdot d(\frac{1}{2}\mu_0\vec{H}^2) + V\vec{H}\cdot d\vec{m}$

一般形式：$đW = \sum_i Y_i\,dy_i$.

Y_i为广义力，y_i为广义坐标。

e.g. $Y_i = (-P, \mu_0 H, E, \sigma)$

$y_i = (V, m, P, A)$

1.6.

热力学第一定律：

系统与外界传递能量的方式：

(1) 做功，外参量改变。

(2) 传递热量。

绝热过程：系统状态只由(1)改变。

系统经绝热过程做功只由初态、末态决定，与过程无关。

↳ 定义内能 U，$W = U_B - U_A$

非绝热过程：定义热量 $Q = U_B - U_A - W$

$\Rightarrow$ 热力学第一定律：$U_B - U_A = W + Q$，

$$dU = đW + đQ$$

1.7、热容量与焓

$$C_y = \frac{đQ_y}{dT}$$ 等容：$W=0$，$đQ = dU$

$\hookrightarrow$ $$C_V = \frac{đQ}{dT}\Big|_V = \frac{dU}{dT} = \left(\frac{\partial U}{\partial T}\right)_V$$

$$C_P = \frac{đQ}{dT}\Big|_P = \frac{dU + PdV}{dT}\Big|_P = \left(\frac{\partial U}{\partial T}\right)_P + P\left(\frac{\partial V}{\partial T}\right)_P$$

定义焓：$H = U + PV$

$$C_P = \left(\frac{\partial H}{\partial T}\right)_T$$

1.8、理想气体的性质

1. 内能、焓

$$U = U(T,V) = U(T) \quad (\text{理想气体})$$

$$PV = nRT$$

$$H = U + PV = H(T)$$

$$C_V = \left(\frac{\partial U}{\partial T}\right)_V = \frac{dU}{dT} = C_V(T)$$

$$C_P = \left(\frac{\partial H}{\partial T}\right)_P = \frac{dH}{dT} = C_P(T)$$

$$C_P - C_V = \frac{d}{dT}(H-U) = \frac{d(PV)}{dT} = nR$$

$$\gamma(T) = \frac{C_P}{C_V} \rightarrow C_V = \frac{nR}{\gamma-1},\quad C_P = \frac{\gamma nR}{\gamma-1}$$

$$U = U_0 + \int C_V(T)\,dT$$

$$H = H_0 + \int C_P(T)\,dT$$

2. 绝热方程

$$dU = đW = -PdV$$

$$nR\,dT = PdV + VdP$$

$\rightarrow$ $$-PdV = dU = C_V dT = \frac{nR}{\gamma-1}\cdot\frac{1}{nR}(PdV + VdP)$$

$\rightarrow$ $$\frac{1}{\gamma-1}\left(\gamma\frac{dV}{V} + \frac{dP}{P}\right) = 0$$

$\Rightarrow$ $PV^\gamma = \text{Const.}$

$\hookrightarrow$ $TV^{\gamma-1} = \text{Const.}$，$P^{\gamma-1}T^{-\gamma} = \text{Const.}$

声速测定.

$$a = \sqrt{\left(\frac{\partial P}{\partial \rho}\right)_s} = \sqrt{\left(\frac{\partial P}{\partial V}\right)_\rho \left(\frac{\partial V}{\partial \rho}\right)_s}$$

比体积 $v = \frac{1}{\rho}$, $\left(\frac{\partial v}{\partial \rho}\right)_s = -\frac{d\rho}{\rho^2} = -v^2 d\rho$

$$a = \sqrt{-v^2 \left(\frac{\partial P}{\partial v}\right)_s}$$

$$0 = d(PV^\gamma) = V^\gamma dP + \gamma P V^{\gamma-1} dV \to \left(\frac{\partial P}{\partial V}\right)_s = -\gamma \frac{P}{V}$$

$$a = \sqrt{\gamma p V} = \sqrt{\gamma \frac{P}{\rho}}$$

3. 能量方程式

$$u = u(T, v)$$

$$du = \left(\frac{\partial u}{\partial T}\right)_v dT + \left(\frac{\partial u}{\partial v}\right)_T dv = C_v dT + \left(\frac{\partial u}{\partial v}\right)_T dv$$

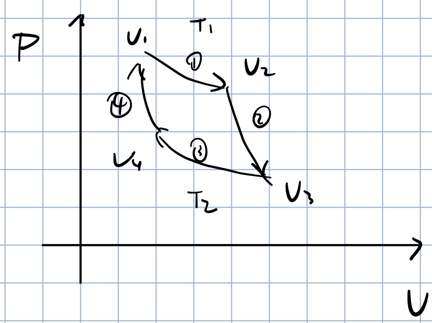
$$C_p = \left(\frac{dQ}{dT}\right)_P = \left(\frac{du + Pdv}{dT}\right)_P = P\left(\frac{\partial v}{\partial T}\right)_P + C_v + \left(\frac{\partial u}{\partial v}\right)_T \left(\frac{\partial v}{\partial T}\right)_P$$

$$du = C_v dT + \left[(C_p - C_v)\left(\frac{\partial T}{\partial v}\right)_p - P\right] dv$$

1.9. 理想气体的卡诺循环.

循环效率 $\eta = \frac{W}{Q_{吸}} = 1 - \frac{T_2}{T_1} < 1$

制冷效率 $\varepsilon = \frac{Q_2}{W} = \frac{T_2}{T_1 - T_2}$



① : $\Delta U = 0$

$$Q_1 = -W_1 = \int_{V_1}^{V_2} p\, dV = \nu R T_1 \ln \frac{V_2}{V_1}$$

② , $\Delta Q = 0$

③ : $\Delta U = 0$

$$Q_2 = -W_2 = \int_{V_3}^{V_4} p\, dV = \nu R T \ln \frac{V_4}{V_3}$$

④ : $\Delta Q = 0$

$$\eta = \frac{W}{Q_{吸}} = \frac{Q_1 + Q_2}{Q_1} = 1 - \frac{\ln V_3/V_4}{\ln V_2/V_1} \cdot \frac{T_2}{T_1}$$

由 $PV^\gamma = \text{Const.} \to TV^{\gamma-1} = \text{Const.}$

$$\begin{cases} T_1 V_1^{\gamma-1} = T_2 V_4^{\gamma-1} \\ T_1 V_2^{\gamma-1} = T_2 V_3^{\gamma-1} \end{cases} \Rightarrow \frac{V_1}{V_4} = \frac{V_2}{V_3} = \left(\frac{T_2}{T_1}\right)^{\frac{1}{\gamma-1}}, \quad \frac{V_2}{V_1} = \frac{V_3}{V_4}$$

$$\Rightarrow \eta = 1 - \frac{T_2}{T_1}$$

1.10 热力学第二定律

开尔文表述：不可能从单一热源吸热，使之变为

1.11 热力学第二定律数学表达式、熵

卡诺定理：

a) 工作在两个一定温度之间的热机，可逆热机效率 max

b) 所有工作在一定温度之间的可逆～，效率相同。

$$\eta = 1 - \frac{Q_2}{Q_1} \leq 1 - \frac{T_2}{T_1} \rightarrow \frac{Q_1}{T_1} - \frac{Q_2}{T_2} \leq 0$$

定义 Q 为吸热，$\sum_i \frac{Q_i}{T_i} \leq 0 \qquad \oint \frac{đQ}{T} \leq 0$

对可逆过程 R, R' $(A \leftrightarrow B)$

$$\int_A^B \frac{đQ_R}{T} + \int_B^A \frac{đQ_{R'}}{T} = 0$$

↳ 定义态函数，熵：$S_B - S_A = \int_A^B \frac{đQ}{T}$

对不可逆过程 IR：

$$\int_P^{P_0} \frac{đQ_R}{T} + \int_{P_0}^{P} \frac{đQ_{IR}}{T} < 0$$

$$\Delta S = S - S_0 = \int_{P_0}^{P} \frac{đQ_R}{T} > \int_{P_0}^{P} \frac{đQ_{IR}}{T}$$

$$dS \geq \frac{đQ}{T}$$

热力学的基本微分方程：

$$dU = dQ + đW = TdS - PdV$$

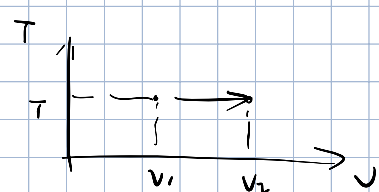
$$= TdS + \sum_i (Y_i\, dy_i + \underbrace{\mu_i\, dN_i}_{\text{开放系统}})$$

1.12. 熵增加原理

$$\Delta S = \int \frac{đQ}{T} \geq 0$$

1.13. 不可逆过程例子

1) 自由膨胀

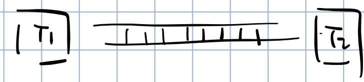


$$\Delta S = \frac{Q}{T} = \frac{\Delta U - W}{T} = \int_{V_1}^{V_2} P\,dV \cdot \frac{1}{T}$$

$$= NR \ln\left(\frac{V_2}{V_1}\right)$$

(2) 初态：$T(x) = T_1 - \frac{T_1-T_2}{L}x$

末态：$T_f = \frac{1}{2}(T_1+T_2)$



$$\Delta[S(x)dx] = \int_i^f \frac{dQ}{T} = c_p\rho A\, dx \int_{T(x)}^{T_f}\frac{dT}{T}$$

$$= c_p\rho A \ln\left(\frac{T_f}{T(x)}\right) dx$$

$$\Delta S = \int_0^L \Delta[S(x)dx]$$

$$= c_p\rho A\int_0^L \ln\frac{\frac{1}{2}(T_1+T_2)}{T_1-\frac{T_1-T_2}{L}x}\,dx$$

$$= c_p\rho A\left\{\ln\frac{1}{2}(T_1+T_2)\cdot L + \frac{L}{T_1-T_2}\left[\left(T_1-\frac{T_1-T_2}{L}x\right)\ln\left(T_1-\frac{T_1-T_2}{L}x\right) - T_1 + \frac{T_1-T_2}{L}x\Big|_0^L\right]\right\}$$

$$= LAc_p\rho\left[1+\ln\left(\frac{T_1+T_2}{2}\right) + \frac{T_1\ln T_1 - T_2\ln T_2}{T_1-T_2}\right]$$

$$= LAc_p\rho\sum_{m=1}^{\infty}\frac{1}{2m(2m+1)}\left(\frac{T_1-T_2}{T_1+T_2}\right)^{2m} > 0$$

(3) 两杯水混合 $T_1, T_2 \to \frac{1}{2}(T_1+T_2)$

为什么是等压？

$$dS = \frac{dQ}{T} = \frac{dU+PdV}{T} = \frac{dH}{T} = c_p\frac{dT}{T}$$

$$\Delta S = \Delta S_1 + \Delta S_2 = c_p\cdot\left[\ln\left(\frac{T_1+T_2}{2T_1}\right) + \ln\left(\frac{T_1+T_2}{2T_2}\right)\right]$$

$$= c_p \ln\frac{(T_1+T_2)^2}{4T_1T_2} \geq 0$$

1.14. 自由能、Gibbs函数.

$$S_B - S_A \geq \frac{Q}{T} = \frac{U_B-U_A-W}{T}$$

定义态函数，自由能 $F = U - TS$

$$F_A - F_B \geq -W$$

最大功定理：对外做最大功 ≤ 自由能减少.

在等温，等压过程中：

外界对系统做其它功

$$S_B - S_A \geq \frac{U_B-U_A-W}{T}, \quad W = -P(V_B-V_A) + W_1$$

$$\geq \frac{U_B-U_A+P(V_B-V_A)-W_1}{T}$$

定义态函数，Gibbs自由能 $G = U - TS + PV$

$$G_A - G_B \geq -W_1$$

在等p,T过程中，除pdV功外，可做最大功.